

CAR: Collision-Avoidance Retargeting for Varied Skeletal Architectures

Yu Cao
Ant Group
China
caoyu0708@foxmail.com

MingHui Yang
Ant Group
China
minghui.ymh@antgroup.com

Abstract

Motion retargeting is essential for digital human and animation applications, typically demands strict skeletal structure adherence, often simplifying Mixamo-based skeletons to focus on action semantics rather than geometric differences. This paper introduces Collision-Avoidance Retargeting(CAR) method that does not rely on data-driven approaches and can be applicable to different skeleton architectures in cold start situations, ensuring the integrity of the drivable skeleton of the target character and reducing the occurrence of mesh interpenetrations due to shape geometric differences. The method has two stages. Stage one aligns bone chains and transforms motions with matrix operations. Stage two uses geometric data to refine joint rotations, preventing collisions and preserving action accuracy. CAR outperforms existing methods, as shown in extensive experiments, in both quantitative metrics and qualitative assessments.

CCS Concepts

• **Computing methodologies** → **Motion processing; Collision detection.**

Keywords

Motion retargeting, Skeleton animation

ACM Reference Format:

Yu Cao and MingHui Yang. 2024. CAR: Collision-Avoidance Retargeting for Varied Skeletal Architectures. In *SIGGRAPH Asia 2024 Technical Communications (SA Technical Communications '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3681758.3698007>

1 Introduction

Character animation is essential in 3D gaming and animation, typically created through manual or motion capture methods, both of which are costly and character-specific. Motion retargeting is key for transferring animations across characters with different skeletal structures, but traditional methods are limited by the need for manual correspondence and are prone to errors. Neural network-based approaches, while promising, struggle with cold-start issues and lack of paired training data.



This work is licensed under a Creative Commons Attribution International 4.0 License.

SA Technical Communications '24, December 03–06, 2024, Tokyo, Japan
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1140-4/24/12
<https://doi.org/10.1145/3681758.3698007>

This paper introduces the Collision-Avoidance Retargeting(CAR), a two-stage process that addresses these challenges. Stage one focuses on skeletal-level motion transfer, using heuristic strategies to align bone chains and transfer motion data without altering the original file structure. Stage two introduces shape geometry to minimize interpenetration, employing an improved Signed Distance Field (SDF) and a normalized distance matrix to guide optimization and preserve motion semantics. The contributions of this study are as follows:

- (1) We propose a motion retargeting method suitable for various humanoid skeletons, eliminating the need for paired motion data, and enabling automatic alignment of motion semantics at the skeletal level along with reasonable geometric deformation.
- (2) We adopt a data processing approach well-suited for practical animation production, avoiding modifications to the original skeletal structure and preventing irreversible changes to character data.
- (3) Our method demonstrates superior performance on public datasets and is effective for characters with diverse skeletal architectures, increasing its potential for broad application.

2 Related Work

Motion retargeting has been extensively studied, with early works framing it as a constrained optimization problem. Gleicher [Gleicher 1998] pioneered kinematic constraint-based optimization of motion sequences, while subsequent research integrated physical plausibility to guide the optimization process [Al Borno et al. 2018; Popović and Witkin 1999]. With the rise of deep learning, scholars have explored data-driven approaches. Jang et al. [Jang et al. 2018] utilized deep learning with paired data for human motion generation, while Villegas et al [Villegas et al. 2018] employed unpaired data and unsupervised learning for motion retargeting. Recent methods have incorporated skeletal and shape information, such as Aberman et al.'s SAN [Aberman et al. 2020] and Zhang's R²ET [Zhang et al. 2023], which aim to preserve motion semantics while avoiding interpenetration.

This paper contributes to the field by proposing a novel two-stage method that does not rely on data-driven approaches, addressing limitations in cold-start scenarios and generalization. Our method leverages heuristic and optimization strategies to transfer motion accurately while considering both skeletal and geometric levels, demonstrating higher versatility and practical applicability across various skeletal architectures.

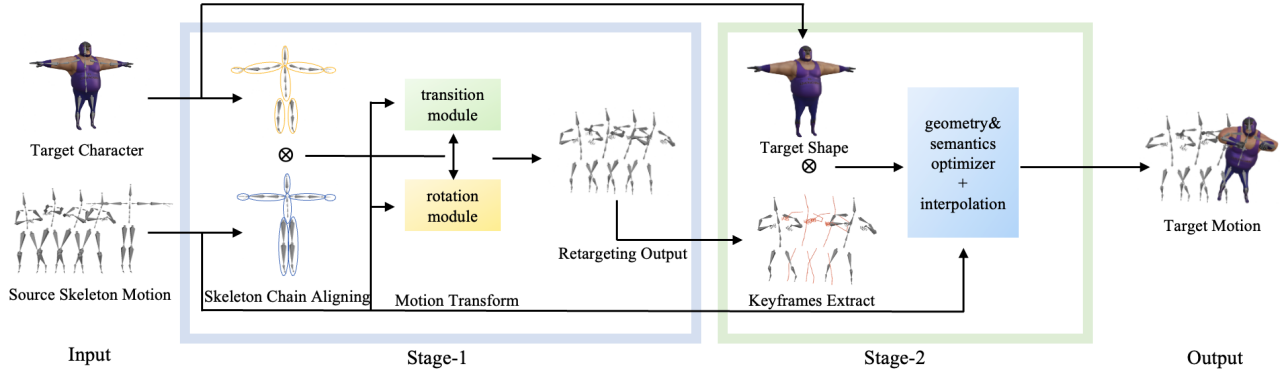


Figure 1: Method overview. The CAR method takes a target character model and motion data from a source skeleton as input, and produces motion data for the target character.

3 Methodology

Our two-stage CAR framework, as shown in Figure 1, efficiently transfers motion between characters with minimal interpenetration. Stage one achieves skeletal motion transfer via bone chain alignment, leveraging humanoid structure similarities; rotation transformation to match joint postures; and translation adjustment to fit the target’s proportions. Stage two refines motion by optimizing keyframes and interpolating for smooth transitions, addressing initial interpenetration. This approach ensures seamless motion adaptation without altering the target character’s structure.

3.1 Skeletal-Based Motion Transfer

Character animation relies on skeletal motion, and our framework’s first stage focuses on transferring this motion from the source to the target character by leveraging their skeletal data. We’ve developed a heuristic-based approach that minimizes the need for extensive data and accommodates diverse skeletal architectures. Here are the specifics of our method:

Bone Chain Alignment. The primary objective of motion retargeting tasks is to accurately transfer motion from source to target skeleton joints. Creating joint correspondences between source and target skeletons is crucial, but constructing them can be difficult and usually requires manual work. This challenge limits the broader use of motion retargeting methods due to the demands of joint matching.

A bone chain refers to a series of bones connected in sequence, commonly used in inverse kinematics. Finding corresponding bone chains on different skeletons is much easier than finding corresponding joints. Our method defines four bone chains, represented as lists of joints, starting from the root joint and ending at the left hand, right hand, left foot, and right foot, respectively. The number of joints included in each bone chain can vary, but the names of the start and end joints need to be similar because we determine the ends of the chain based on the textual similarity.

Rotation Transformation Module. In FBX file instance, for any joint J , we can retrieve its global rotation matrix R_w and global translation vector t_w with respect to the world coordinate system. Additionally, we can access its local rotation matrix R_l and local

translation vector t_l , both relative to the coordinate system of its parent joint. The local rotation consists of the joint’s pre-rotation R_{pre} which remains constant over time, and the real-time rotation R_r . Except for the root joint, the local translation of other joints is typically fixed, indicative of the skeleton’s rigidity.

It is noteworthy that the joint pre-rotation is defined relative to the coordinate system of the parent joint, whereas the real-time rotation is defined with respect to an intermediate coordinate system that has been pre-rotated by the parent joint’s coordinates. Consequently, this is manifested as a matrix post-multiplication. The aforementioned rotation matrices adhere to the following relationship:

$$R_w = R_{w0}R_{l1} \dots R_l = R_{w0}R_{pre0}R_{r0} \dots R_{pre}R_r \quad (1)$$

The rotation transformation module is designed to transfer rotations from the aligned bone chain pairs of the source skeleton to the target skeleton. Assuming the source bone chain consists of n joints $J_0, \dots, J_n$ and the target bone chain comprises m joints $J_{t0}, \dots, J_{tm}$, the specific methodology is as follows:

- (1) We obtain the following matrices for each joint of the two characters in their initial poses: joint pre-rotation, global rotation, and an intermediate variable between real-time rotation and global rotation, which we refer to as the base rotation R_{base} , denoted as $R_{w0}R_{pre0}R_{r0} \dots R_{pre}$.
- (2) For the source skeleton, the joint’s real-time rotation is converted into a rotation relative to the world coordinate system, denoted as S , and is represented as: $S = R_{base}R_rR_{base}^{-1}$.
- (3) The rotation of the source bone chain can be expressed as $R_{chain_i} = S_i \dots S_0$, where i is in the range 0 to n . Assuming the rotation of the target bone chain is $R_{t_chain_j}$, where j is in the range 0 to m , we define a piecewise function to calculate the rotation S_{tj} for each joint in the target bone chain, where Slerp (Spherical Linear Interpolation) is a spherical linear interpolation algorithm:

$$\begin{aligned} R_{t_chain_0} &= R_{chain_0} \\ R_{t_chain_j} &= \text{Slerp}(I, S_{i+1}, (\frac{j}{m} - \frac{i}{n})R_{chain_i}) \end{aligned} \quad (2)$$

- (4) Finally, the rotation S of the target bone chain joints relative to the world coordinate system is converted into the real-time rotation R_{tr} .

Assign the computed real-time rotation R_{tr} to the target skeleton to complete the posture transfer from the source skeleton.

Translation Transformation Module. In addition to rotations, root joint translations are critical for character motion. However, differences in character proportions require these translations to be scaled. We retrieve global translation curves for the root joint and lower body endpoints, scale them using the leg length ratio, and average the deduced root joint curves. The least squares method optimizes this scaling factor, ensuring the source's scaled root joint curve closely matches the average curve. This optimal scaling ratio is then applied to estimate the target's global translations, completing the motion transfer.

By integrating these modules, stage one enables the redirection of skeletal animations between different characters with minimal prior knowledge and simple operations. Utilizing the Autodesk[Autodesk [n. d.]] environment, our approach preserves the original skeletal structures, demonstrating adaptability to a wide range of skeletal diversity.

3.2 Geometry-Based Motion Correction

Stage two focuses on geometry-based motion correction to address potential interpenetration issues that may occur due to differences in character geometry after skeletal motion retargeting. We firstly identify keyframes in the animation, then define a set of losses, and fine-tune the motion by optimizing these losses to ensure the transferred motion is geometrically plausible.

Keyframe Extraction. Optimization is computationally expensive, making it impractical to apply to every frame. We automate the extraction of keyframes that likely require optimization, based on joint distance variations and penetration severity. This strategy is inspired by human practices in motion correction, as detailed in Algorithm 1.

Interpenetration Loss. To quantify interpenetration, we employ the Signed Distance Field (SDF), and define positive values indicate proximity to boundaries, and values outside the scene are zero. Specifically, we voxelize the vertices of the character's head, torso, and limbs into a voxel grid of size $N \times N \times N$ to create separate SDFs for each body part. We then combine the SDFs of these body parts to create a comprehensive distance field, denoted as Φ . The interpenetration loss is defined as follows:

$$loss_p = \frac{1}{N_a} \sum_{i \in Limbs} \sum_{v \in V} (\Phi - \Phi_i)(v) \quad (3)$$

where N_a represents the number of body parts for which interpenetration loss is computed, $Limbs$ represents the body parts for which interpenetration loss is computed, V represents all vertices.

Motion Semantic Loss. To preserve the original motion semantics during correction, we also model the semantic features. Firstly, a normalized joint distance matrix D is used to represent the semantic features of the original motion. We define the motion similarity

Algorithm 1 Keyframe extraction algorithm

Input: Motion sequence, F

Output: Keyframes, K

```

1: Obtain joints global position.
2: Calculate joints shortest distance  $joint\_dis$ .
3:  $indices = \text{get\_peaks}(joint\_dis) + \text{get\_troughs}(joint\_dis)$ 
4: for  $i$  in  $indices$  do
5:   Calculate interpenetration in frame  $F(i)$ ,  $p_i$ .
6:   if  $p_i$  is an extremum point and  $p_i' < \text{threshold}$  then
7:      $K \leftarrow F(i)$ 
8:   end if
9: end for
10: return  $K$ 
```

loss as:

$$loss_{sem} = \|\eta(\frac{D_s}{h_s}) - \eta(\frac{D_t}{h_t})\|_2 \quad (4)$$

where $\eta(\cdot)$ is an L1 normalization function, h is the height of the source and target character.

We then impose constraints on the degree of incremental rotation to avoid significant changes in the motion. This is achieved by constructing an incremental rotation constraint loss to limit the magnitude of the rotation.

$$loss_d = \|\theta(\Delta q)\|_2 \quad (5)$$

where $\theta(\cdot)$ is used to obtain the rotation angle of the quaternion Δq .

Additionally, we also construct a joint position loss to further restrict the risk of excessive changes in the motion. Although there will inevitably be differences in joint position after the correction, the effect can be balanced by adjusting hyperparameters. The joint position loss is defined as follows:

$$loss_{pos} = \|W(\hat{P} - P)\|_2 \quad (6)$$

where W is the weight coefficient for the joints, P and $\hat{P}$ represent the joint positions before and after correction, respectively. In practice, applying higher weights to the endpoints can yield better results.

Therefore, the overall motion semantic loss can be defined as follows:

$$loss = loss_p + \lambda_1 loss_{sem} + \lambda_2 loss_d + \lambda_3 loss_{pos} \quad (7)$$

where $\lambda_1, \lambda_2, \lambda_3$ is hyperparameters.

We employ the BFGS method to optimize the loss and adjust joint rotations on keyframes, followed by interpolation for frames in between. Incremental rotations are relative to the world coordinate system, which simplifies the process by avoiding discrepancies in the local coordinate systems

4 Experiments

Datasets. We use Mixamo dataset [Adobe [n. d.]], a widely recognized resource providing over 2400 motion sequences for 108 characters. To construct our evaluation set, we randomly selected 50 motion sequences using the "XBot" character as the source and eight additional characters as targets. Acknowledging potential

Table 1: Comparison on Mixamo dataset. We apply MPJPE and MPJPE^{lc} to evaluate the accuracy (multiplied by 10⁴). Pen means interpenetration value.

Methods	MPJPE↓	MPJPE ^{lc} ↓	Pen↓
GT	-	-	9.22
Copy	238	0.959	9.22
SAN	17.674	15.173	15.482
PAN	15.086	12.129	12.197
R ² ET	20.205	3.378	7.177
CAR _{w/oG}	10.13	0.712	9.077
CAR(Ours)	11.1	1.125	3.922

data inconsistencies in Mixamo, we also calculated ground truth evaluations for reference.

Evaluation Metrics. We assessed our method using both skeletal and geometric perspectives. Global and local Mean Joint Position Error (MPJPE)[Aberman et al. 2020] metrics were employed to evaluate the accuracy of joint positions post-retargeting against the ground truth. Specifically, we calculated MPJPE using 22 joints, excluding the fingers, as the baseline methods have varying treatments for the fingers. At the geometric level, we calculate the penetration loss(Pen) using the formula(3), which represents the average degree of interpenetration of the limbs in the motion sequence.

Quantitative Results. The quantitative comparison is shown in Table 1. Quantitative comparisons demonstrated CAR’s superiority over state-of-the-art methods. Notably, CAR_{w/oG} results exhibited lower local MPJPE than directly copying source motion data, underscoring accurate motion transfer while preserving original semantics. Directly copying the root joint translation on characters of different heights significantly affects the joint positions throughout the body, resulting in a large MPJPE for the Copy method.

Both SAN and PAN are skeletal retargeting methods that neglect the influence of shape geometry, leading to increased instances of penetration in generated motions and higher penetration loss metrics. R²ET mitigates this issue by incorporating a shape-aware module. CAR_{w/oG} also disregard shape geometry, exhibit similar penetration levels compared to the copying method. CAR significantly reduces penetration, while there is a slight increase in MPJPE from CAR_{w/oG}. However, it must be noted that our method optimized the penetration loss in stage two, hence its better performance on this metric, which necessitates further qualitative analysis.

Qualitative Results. The Figure 2 shows CAR’s ability to restore motion semantics accurately and diminish interpenetration. Compared to methods neglecting shape geometry, our results were more stable and smooth, indicating effective motion correction. The R²ET method, while stable, showed some jittering artifacts, whereas our method provided more natural motion transitions. Beyond Mixamo characters, as shown in Figure 3, we validated CAR’s compatibility with diverse skeletons, including cartoon and realistic characters. Unlike comparison methods that require target skeleton alteration, CAR achieved motion retargeting without modifying the target’s skeleton structure, showcasing its generality.

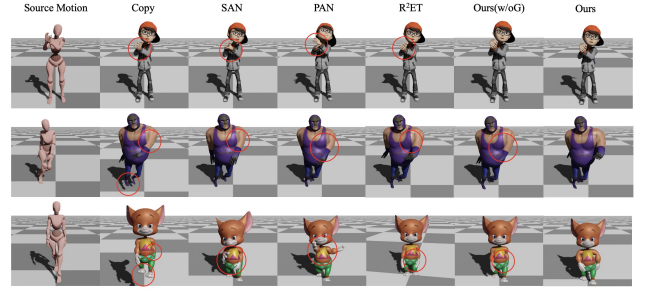


Figure 2: Results on Mixamo characters.

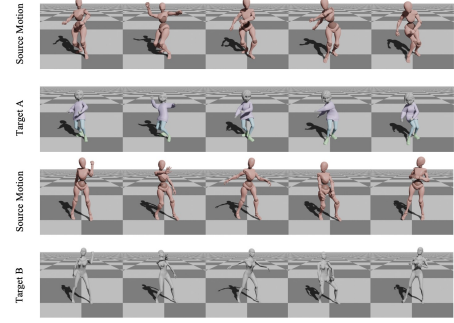


Figure 3: Results on Non-Mixamo characters.

Limitations. The CAR method also has two limitations. The first stage can perform automated skeleton alignment, but it requires that both the source and target skeletons be humanoid and have reasonably named joints. The second stage need much more time to execute the optimization process than neural network-based inference methods. Our executions on the Xeon E5 take an average of approximately 3 seconds.

5 Conclusions

We introduce the Collision-Avoidance Retargeting method (CAR), a two-stage approach that integrates skeleton-based motion transfer with geometry-based correction. This method outperforms existing methods both quantitatively and qualitatively. It effectively transfers motion while minimizing interpenetration across various humanoid skeletal architectures. Future work will integrate inverse kinematics, enhance ground contact constraints for realism, and broaden the motion correction scope.

References

- Kfir Aberman, Peizhuo Li, Dani Lischinski, Olga Sorkine-Hornung, Daniel Cohen-Or, and Baoquan Chen. 2020. Skeleton-aware networks for deep motion retargeting. *ACM Transactions on Graphics (TOG)* 39, 4 (2020), 62–1.
- Adobe. [n. d.]. Mixamo. <https://www.mixamo.com/>. Accessed: 2023-10-01.
- Mazen Al Borno, Ludovic Righetti, Michael J Black, Scott L Delp, Eugene Fiume, and Javier Romero. 2018. Robust Physics-based Motion Retargeting with Realistic Body Shapes. In *Computer Graphics Forum*, Vol. 37. Wiley Online Library, 81–92.
- Autodesk. [n. d.]. FBX SDK. <https://aps.autodesk.com/developer/overview/fbx-sdk>.
- Michael L. Gleicher. 1998. Retargetting motion to new characters. (1998).
- Hanyoung Jang, Byungjun Kwon, Moonwon Yu, Seong Uk Kim, and Jongmin Kim. 2018. A variational u-net for motion retargeting. In *SIGGRAPH Asia 2018 Posters*. 1–2.

- Zoran Popović and Andrew Witkin. 1999. Physically based motion transformation. In *Proceedings of the 26th annual conference on Computer graphics and interactive techniques*. 11–20.
- Ruben Villegas, Jimei Yang, Duygu Ceylan, and Honglak Lee. 2018. Neural kinematic networks for unsupervised motion retargeting. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 8639–8648.
- Jiaxu Zhang, Junwu Weng, Di Kang, Fang Zhao, Shaoli Huang, Xuefei Zhe, Linchao Bao, Ying Shan, Jue Wang, and Zhigang Tu. 2023. Skinned Motion Retargeting with Residual Perception of Motion Semantics & Geometry. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13864–13872.